

Experiment (1) - Report sheet

A) Preparation of 250 mL of 0.20 M solution of a strong acid:

Name of acid: Sulfuric Acid..... Molecular Formula: H₂SO₄.....

Formula weight: 98 g/mol..... Purity: 96%..... Density = 1.84 g/mL

Calculations: [1] $\#n(\text{H}_2\text{SO}_4) = M \times V = 0.2 \times 250 \times 10^{-3} = 0.05 \text{ mole}$

[2] $\text{mass}(\text{H}_2\text{SO}_4) [\text{pure}] = \#n \times \text{MW} = 0.05 \times 98 = 4.9 \text{ g}$

[3] $100 \text{ g stock} \rightarrow 98 \text{ g pure HCl}$
 $X \rightarrow 4.9 \text{ g}$
 $\text{mass stock H}_2\text{SO}_4 = 5 \text{ g}$
 [4] $\text{Density} = \frac{\text{mass of stock}}{\text{volume of stock}}$
 $\text{volume of stock} = \frac{\text{mass}}{d} = \frac{5}{1.84} = 2.7174 \text{ mL}$

B) Preparation of 250 mL of 0.20 M solution of a strong base:

Name of base: potassium hydroxide..... Molecular Formula: KOH.....

Formula weight: 56.11..... Purity: 85%.....

Calculations: [1] $\#n = M \times V = 0.2 \times 250 \times 10^{-3} = 0.05$

[2] $\text{mass}(\text{KOH}) \text{ pure} = \#n \times \text{MW} = 0.05 \times 56.11 = 2.8055 \text{ g}$

[3] $100 \text{ g stock} \rightarrow 85 \text{ g pure}$
 $X \rightarrow 2.8055$
 $\text{mass stock} = 3.3 \text{ g}$

C) Preparation of 100 mL calibration solutions of 1.00, 5.00, 10.0, and 25.0 ppm of element:

Assigned element: Na⁺..... M.W. = 23

Name of compound: Na₃PO₄..... Formula weight: 380.12..... Purity: 97%

Calculations:

- The weight of **compound** required to prepare 250 mL of 100 ppm of element stock solution:

- The volume of 100 ppm element stock solution required to prepare each of the following:

250 mL of 1.00 ppm of element solution: 2.5 mL

250 mL of 5.00 ppm of element solution: 12.5 mL

250 mL of 10.0 ppm of element solution: 25 mL

250 mL of 1.00 ppm of element solution: 2.5 g

$$C_1 V_1 = C_2 V_2$$

$$100 \times V_1 = 1$$

another way :

$$A \quad C_1 V_1 = C_2 V_2 \quad (\text{stock}) \quad (\text{Diluted})$$

$$C_1 V_1 = 0.2 \times 250 \text{ ml}$$

!! ملاحظة C_1 V_1 C_2 V_2
 تركيز المحلول الأصلي C_1
 تركيز المحلول المخفف C_2
 حجم المحلول الأصلي V_1
 حجم المحلول المخفف V_2
 Volume of stock

$$II \text{ Density} = \frac{\text{mass stock}}{\text{Volume stock}}$$

$$1.84 = \frac{\text{mass stock}}{1000 \text{ mL}}$$

$$\text{mass stock} = 1840 \text{ g}$$

$$100 \text{ g stock} \rightarrow 96 \text{ g pure}$$

$$1840 \rightarrow X$$

$$X = 1766.4 \text{ g}$$

mass pure

$$\#n_{\text{pure}} = \frac{\text{mass pure}}{M.W}$$

$$= \frac{1766.4}{98}$$

$$= 18.02 \text{ mole}$$

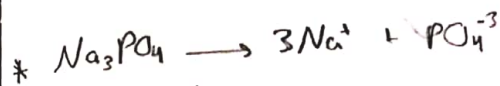
$$M = \frac{\#n}{V} = \frac{18.02}{1} = 18.02$$

$$C_1 V_1 = C_2 V_2$$

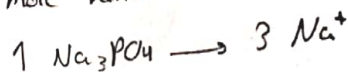
$$V_{\text{stock}} = \frac{0.2 \times 250}{18.02} = 2.775 \text{ ml}$$

□ prepare 250 ml of ppm of Na^+ : $\frac{100}{1000}$

$$\text{mass} = M \times V = \frac{250}{1000} \times \frac{100 \text{ mg}}{1} = 25 \text{ mg of } \text{Na}^+$$

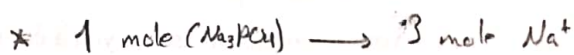


mole ratio



$$\text{mass of } \text{Na}^+ = 25 \text{ mg} \times 1000 = 25 \times 10^3 \text{ g}$$

$$\text{moles of } \text{Na}^+ = \frac{\text{mass}}{M.W} = \frac{25 \times 10^3}{23} = 1086.96 \text{ mole}$$



$$X \rightarrow 1086.96$$

$$\text{moles of } \text{Na}_3\text{PO}_4 = 362.32$$

$$\text{mass } \text{Na}_3\text{PO}_4 = 362.32 \times 380.12 \text{ (MW)} = 137724.6377 \text{ g}$$

$$\text{mass stock} = \frac{100}{97} \times 137724.6377 \text{ g} = 141984.1626 \text{ g}$$

OR

$$\text{mass } \text{Na}_3\text{PO}_4 \text{ stock} = \text{mass } \text{Na}^+ \times \left[\frac{M.W \text{ Na}_3\text{PO}_4}{M.W \text{ Na}^+} \times \text{mole ratio} \right] \times \frac{100}{97}$$

$$= 25 \times 10^3 \left[\frac{380.12}{23} \times \frac{1}{3} \right] \times \frac{100}{97}$$

$$= 141984.1626 \text{ g}$$

$$-\frac{1}{2} = 0.1419 \text{ g}$$

9

N = 19.5

في 6 beakers
المجموع الكلي هو 119.5
المادة هي 25 ml وزايم
منه على 1 balance و 2 balance
في احوال اشفاد ما يتبع مقبول
هل هي دقيقة؟ طبعاً
Accuracy precision
طبياً لا يجب ان يتم الشطب

Experiment (2) - Report sheet

Beaker	Wt. on balance 1	Wt. on balance 2
1	19.0029	19.0003
2	19.1490	19.1465
3	19.2328	19.2300
4	19.5611	19.5583
5	19.931	19.9302
6	18.9794	18.9764
Average $\bar{x}$	19.2094	19.221
Standard deviation S	0.218	0.223
Coefficient of variation CV	1.134	1.16
Confidence interval	(19.43, 12.98)	(19.433, 12.977)

t (tabulated) (95%) = 2.571

Case 1: Comparing measured results with a "True" value

True value (μ): 19.5 t_1 (calculated) = 3.27 t_2 (calculated) = 3.07

Conclusion:

There is no significant difference between the true value (μ) and ($\bar{x}_1$) → accurate

There is no significant difference between the true value (μ) and ($\bar{x}_2$) → accurate

Case 2: Comparing replicate measurements

S_{pooled} = 0.0473 t (calculated) = 0.1098 t (tabulated) (95%) = 2.228

Conclusion:

There is no significant difference between $\bar{x}_1$ and $\bar{x}_2$

Rejection of bad data (Q - Test):

The suspect value: 18.9794 $Q_{calculated}$: 0.04 Q_{table} : 0.56

Conclusion:

The suspect value should be rejected/ retained

Case 3: Comparing individual differences

نطرح الجداولين من بعض

Object	Balance 1	Balance 2	Difference
1	13.6428	13.6395	3.3×10^{-3}
2	28.2934	28.2893	282905.71
3	33.4062	33.4014	4.8×10^{-3}
4	19.1802	19.1755	4.7×10^{-3}
5	7.0249	7.0288	3.9×10^{-3}

Average deviation ($\bar{d} = \sum \frac{d_i}{n}$): 56581.14

Standard deviation (S_d): 26519.277

$$t(\text{calculated}) = \frac{\bar{d}}{S_d} \times \sqrt{n} = 0.99$$

t (tabulated) (95%) = 2.776

Conclusion:

There is/ there is no significant difference between Balance₁ and Balance₂.

$t_{cal} > t_{tab}$: there is sig diff in this result
 $t_{cal} < t_{tab}$ " " no sig diff in this result

This result is no sig diff, it's doesn't need calibration

Experiment (2) - Report sheet

#13
Mass Unknown 0.5812 g

Mass BaSO₄: 0.487 g

Gravimetric factor: 0.3436

% SO₃ in the Unknown: 16.73 %

Calculation:

$$* \text{Mass BaSO}_4 = (1.013) - (0.526) = (0.487) \text{ g}$$

$$* \text{Gravimetric Factor} = \left[\frac{\text{M.W SO}_3}{\text{M.W BaSO}_4} \times \frac{1 \text{ SO}_3}{1 \text{ BaSO}_4} \right]$$
$$= \left[\frac{80.066}{233.038} \times \frac{1}{1} \right] = 0.3436$$

$$* \text{SO}_3 \% = \text{mass (BaSO}_4) \times \text{Gravimetric Factor} \times 100$$
$$= 0.487 \times 0.3436 \times 100$$
$$= 16.73 \%$$

Experiment # 4

Application of acid base titration

Report sheet

Name: Shyma Ibrahim Mohammad

Number: 132243

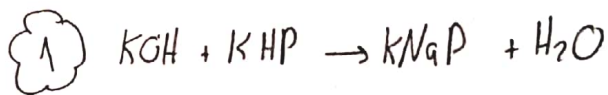
1. Standardization of a NaOH Solution with Potassium Hydrogen Phthalate (KHP)

	Trial 1	Trial 2	Trial 3
Mass of KHP (g)	0.21 g	0.23 g	
Volume of NaOH used (L)	11.7×10^{-3}	14.3×10^{-3}	
Moles of KHP	1.03×10^{-3}	1.13×10^{-3}	
Moles of NaOH	1.13×10^{-3}	1.13×10^{-3}	
Molarity of NaOH	0.088	0.079	
Average Molarity of NaOH	0.084		

2. Determination of the purity of strong acid

Unknown no.: ... H_2SO_4

	Trial 1	Trial 2	Trial 3
Mass of unknown sample (g)	0.33 g	0.33 g	
Molarity of NaOH used	0.084	0.084	
Volume of NaOH used (L)	4.3×10^{-3}	4.6×10^{-3}	
Moles of NaOH	3.61×10^{-4}	3.86×10^{-4}	
Moles of unknown acid	1.81×10^{-4}	1.93×10^{-4}	
H_2SO_4 moles	1.81×10^{-4}	1.93×10^{-4}	
Mass of acid (g)	0.17 g	0.19 g	
Mass percent (%)	51.52 %	57.58 %	
Average Mass percent	54.55 %		



$1 \#n(\text{H}^+) = 1 \#n(\text{OH}^-)$

$\#n(\text{KHP}) = \frac{m}{M.M}$

$= \frac{0.21}{204}$

$= 1.03 \times 10^{-3}$

$\#n(\text{KHP}) = \#n(\text{KOH})$

$1.03 \times 10^{-3} = M.V$

$M = \frac{1.03 \times 10^{-3}}{11.7 \times 10^{-3}}$

$= 0.088$

$\#n(\text{KHP}) = \frac{m}{M.M}$

$= \frac{0.23}{204}$

$= 1.13 \times 10^{-3}$

$\#n(\text{KOH}) = M.V$

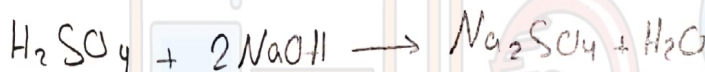
$1.13 \times 10^{-3} = M(14.3 \times 10^{-3})$

$M = 0.079$

Average Molarity $\text{NaOH} = \frac{0.079 + 0.088}{2}$

$= 0.0835 \approx 0.084 \text{ M}$

2



$\rightarrow 2 \#n(\text{H}^+) = 1 \#n(\text{OH}^-) \leftarrow$

$\#n(\text{NaOH}) = M.V$

$= 0.084 \times 4.3 \times 10^{-3}$

$= 3.61 \times 10^{-4}$

$\#n(\text{H}_2\text{SO}_4) = \frac{1}{2} \#n(\text{NaOH})$

$= 1.81 \times 10^{-4}$

total moles $= 1.81 \times 10^{-4} \times \frac{250}{25}$

$= 1.81 \times 10^{-3}$

mass $(\text{H}_2\text{SO}_4) = \text{total moles} \times M.W$

$= 1.81 \times 10^{-3} \times 98$

$= 0.17$

$\text{H}_2\text{SO}_4 \% = \frac{0.17}{0.33} \times 100 = 51.52 \%$

$\#n(\text{NaOH}) = M.V$

$= 0.084 \times 4.6 \times 10^{-3}$

$= 3.86 \times 10^{-4}$

$\#n(\text{H}_2\text{SO}_4) = \frac{1}{2} \#n(\text{NaOH})$

$= 1.93 \times 10^{-4}$

total moles $= 1.93 \times 10^{-4} \times \frac{250}{25}$

$= 1.93 \times 10^{-3}$

mass $(\text{H}_2\text{SO}_4) = \text{total moles} \times M.W$

$= 1.93 \times 10^{-3} \times 98$

$= 0.19$

$\text{H}_2\text{SO}_4 \% = \frac{0.19}{0.33} \times 100$

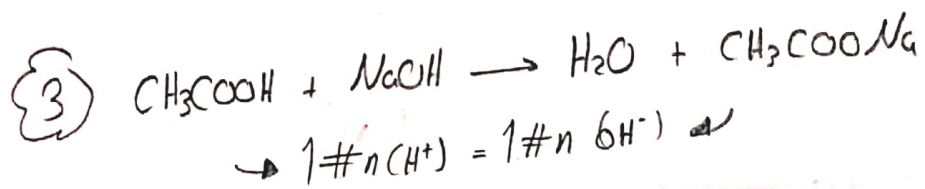
$= 57.58 \%$

3. Determination of total acidity of vinegar:

Unknown no.: ...63.....

	Trial 1	Trial 2	Trial 3
Volume of unknown vinegar (ml)	10 ml		
Molarity of NaOH used	0.084		
Volume of NaOH used (L)	15.1×10^{-3}		
Moles of NaOH	1.27×10^{-3}		
Moles of acetic acid	1.27×10^{-3}		
	calculated moles 0.013		
Mass of acetic acid (g)	0.78 g		
total acidity of vinegar (g acetic acid/ 100 ml vinegar)	7.8×10^{-3}	$0.78 \text{ g}/100 \text{ ml}$	
Average total acidity			

Group



$$\begin{aligned} \#n(\text{NaOH}) &= M V \\ &= 0.084 \times 15.1 \times 10^{-3} \\ &= 1.27 \times 10^{-3} \end{aligned}$$

$$\#n(\text{CH}_3\text{COOH}) = \#n(\text{NaOH}) = 1.27 \times 10^{-3}$$

$$\begin{aligned} \text{total moles} &= 1.27 \times 10^{-3} \times \frac{100}{10} \\ &= 0.013 \end{aligned}$$

$$\begin{aligned} \text{mass of acetic acid} &= \text{total moles} \times \text{MW} \\ &= 0.013 \times 60 \\ &= 0.78 \text{ g} \end{aligned}$$

$$\text{total acidity} = \frac{0.78}{100} = 7.8 \times 10^{-3}$$

Group

Experiment # 5

Application of acid base titration

Report sheet

Name: Shyma Ibrahim Mohammad Number: 132293

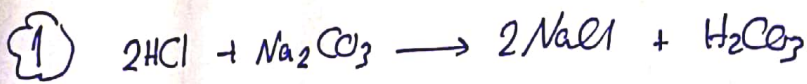
1. Standardization of a HCl Solution with Sodium carbonate (Na_2CO_3)

	Trial 1	Trial 2	Trial 3
Mass of Na_2CO_3 (g)	0.214	0.207	
Moles of Na_2CO_3	2.02×10^{-3}	1.95×10^{-3}	
Moles of HCl	4.04×10^{-3}	3.90×10^{-3}	
Volume of HCl used (L)	35.7×10^{-3}	38.8×10^{-3}	
Molarity of HCl	0.11	0.10	
Average Molarity of HCl	0.105		

2. Determination of the purity of Soda ash

Unknown no.: 7.....

	Trial 1	Trial 2	Trial 3
Mass of unknown sample (g)	1.6044	1.6044	
Molarity of HCl used	0.105		
Volume of HCl used (L)	9.5×10^{-3}	9.7×10^{-3}	
Moles of HCl	9.98×10^{-4}	1.02×10^{-3}	
Moles of Na_2CO_3	4.99×10^{-4}	0.51×10^{-3}	
	Total: 4.99×10^{-3}	5.1×10^{-3}	
Mass of Na_2CO_3 (g)	0.5289	0.5406	
Mass percent (%)	52.66 %	53.82 %	
Average Mass percent	53.24 %		



$$1 \#(\text{H}^+) = 2 \#(\text{OH}^-)$$

trial 1

$$\#(\text{Na}_2\text{CO}_3) = \frac{m}{M \cdot M}$$

$$= \frac{0.214}{106}$$

$$= 2.02 \times 10^{-3}$$

$$\# \text{HCl} = 2 \#(\text{Na}_2\text{CO}_3)$$

$$= 4.04 \times 10^{-3}$$

$$\#(\text{HCl}) = M \cdot V$$

$$\frac{4.04 \times 10^{-3}}{35.7 \times 10^{-3}} = M$$

$$0.11 = M$$

trial 2

$$\#(\text{Na}_2\text{CO}_3) = \frac{0.207}{106}$$

$$= 1.95 \times 10^{-3}$$

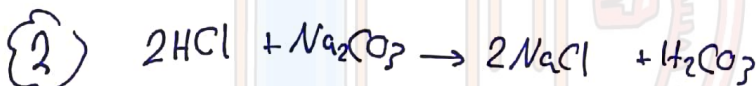
$$\#(\text{HCl}) = 3.90 \times 10^{-3}$$

$$\# \text{HCl} = M \cdot V$$

$$M = \frac{3.90 \times 10^{-3}}{38.8 \times 10^{-3}}$$

$$M = 0.10$$

$$\text{Average } M = \frac{0.11 + 0.10}{2} = 0.105$$



$$\#(\text{H}^+) = 2 \#(\text{OH}^-)$$

trial 1

$$\#(\text{HCl}) = M \cdot V$$

$$= 0.105 \times 9.5 \times 10^{-3}$$

$$= 9.98 \times 10^{-4}$$

$$\#(\text{Na}_2\text{CO}_3) = \frac{1}{2} \#(\text{HCl})$$

$$= 4.99 \times 10^{-4}$$

$$\text{total moles} = 4.99 \times 10^{-4} \times \frac{250}{25}$$

$$= 4.99 \times 10^{-3}$$

$$\text{mass} = \text{total moles} \times 106$$

$$= 0.52894$$

$$\text{wt\%} = \frac{0.5289}{1.0044} \times 100 = 52.66\%$$

trial 2

$$\#(\text{HCl}) = 0.105 \times 9.7 \times 10^{-3}$$

$$= 1.02 \times 10^{-3}$$

$$\#(\text{Na}_2\text{CO}_3) = 0.51 \times 10^{-3}$$

$$\text{total moles} = 0.51 \times 10^{-3} \times \frac{250}{25}$$

$$= 5.1 \times 10^{-3}$$

$$\text{mass} = 0.5406 \text{ g}$$

$$\text{wt\%} = \frac{0.5406}{1.0044} \times 100$$

$$= 53.82\%$$

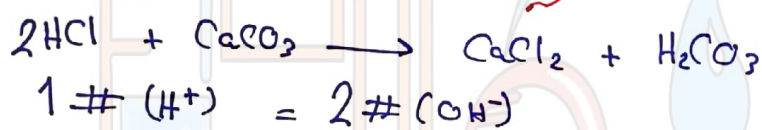
Average mass percent

$$= \frac{52.66 + 53.82}{2}$$

$$= 53.24\%$$

3. Determination of total alkalinity of water:

	Trial 1	Trial 2	Trial 3
Volume of tap water (ml)	50 ml	50 ml	
Molarity of HCl used			
Volume of HCl used (L)	0.105		
Moles of HCl	4.1×10^{-3}	3.5×10^{-3}	
Moles of CaCO_3	4.31×10^{-4}	3.68×10^{-4}	
Mass of CaCO_3 (g)	2.16×10^{-4}	1.84×10^{-4}	
total alkalinity of water (ppm CaCO_3)	21.6×10^{-3}	18.4×10^{-3}	
Average total acidity	432 mg/L	368 mg/L	



Trial 1

$$\#(\text{HCl}) = MV$$

$$= 0.105 \times 4.1 \times 10^{-3}$$

$$= 4.31 \times 10^{-4}$$

$$\#(\text{CaCO}_3) = \frac{1}{2} \#(\text{HCl})$$

$$= 2.16 \times 10^{-4}$$

$$\text{mass}(\text{CaCO}_3) [\text{mg}] = \# \text{CaCO}_3 \times M.M \times 10^3$$

$$= 21.6 \text{ mg}$$

$$\text{ppm} = \frac{\text{mass (mg)}}{V (\text{L})}$$

$$= \frac{21.6}{50 \times 10^{-3}} = 432 \text{ mg/L}$$

Trial 2

$$\#(\text{HCl}) = 0.105 \times 3.5 \times 10^{-3}$$

$$= 3.68 \times 10^{-4}$$

$$\#(\text{CaCO}_3) = 1.84 \times 10^{-4}$$

$$\text{mass} [\text{mg}] = 18.4 \text{ mg}$$

$$\text{ppm} = \frac{18.4}{50 \times 10^{-3}}$$

$$= 368 \text{ mg/L}$$

$$\text{Average} = \frac{368 + 432}{2} = 400 \text{ mg/L}$$

Experiment 6 (Back Titration):
Report sheet

Name: Shyma Ibrahim Mohammed

Number: 132293

Determination of % NH_4Cl in fertilizer

Unknown no... 26...

	Trial 1	Trial 2
Mass of unknown (g)	0.2354 g	
Volume of OH^- added (L)	$30 \times 10^{-3} \text{ L}$	
Molarity of OH^-	0.089 M	
Moles of OH^- added	$2.67 \times 10^{-3} \text{ moles}$	
Volume of HCl back titrated (L)	$7.2 \times 10^{-3} \text{ L}$	
Molarity of HCl	0.103	
Moles of HCl back titrated	$7.416 \times 10^{-4} \text{ moles}$	
Moles of NaOH unreacted	$7.416 \times 10^{-4} \text{ moles}$	
Moles of NaOH reacted	$1.9284 \times 10^{-3} \text{ moles}$	
Moles of NH_4Cl	$1.9284 \times 10^{-3} \text{ moles}$	
Mass of NH_4Cl	0.1032 g	
% NH_4Cl	43.84 %	
Average % NH_4Cl	43.84 %	

mass = 0.2354 g

V = 7.2 ml

M HCl = 0.103

M NaOH = 0.089

Determination of % CaCO_3 in Antacid

Mass of Antacid tablet... 0.2438 g

	Trial 1	Trial 2
Volume of HCl added (L)	$50 \times 10^{-3} \text{ L}$	
Molarity of HCl	0.103 M	
Moles of HCl	$5.15 \times 10^{-3} \text{ moles}$	
Volume of NaOH back titrated	$26 \times 10^{-3} \text{ L}$	
Molarity of NaOH	0.089	
Moles of NaOH back titrated	2.314×10^{-3}	
Mol CaCO_3	$1.418 \times 10^{-3} \text{ mol}$	
Mass of CaCO_3 (g)	0.1418 g	
% CaCO_3	58.16 %	
Average % CaCO_3	58.16 %	

1

$$\text{1] total Moles (NaOH) added} = M \cdot V$$

$$= 0.089 \times 30 \times 10^{-3} = 2.67 \times 10^{-3} \text{ moles}$$

$$\text{2] Mol HCl [Back titrated]} = \text{Mol NaOH (unReacted)}$$

$$= M_{\text{HCl}} \times V_{\text{HCl}} \quad \leftarrow \text{burette volume}$$

$$= 0.103 \times 7.2 \times 10^{-3}$$

$$= 7.416 \times 10^{-4} \text{ moles}$$

$$\text{3] Mol NaOH (reacted)} = (2.67 \times 10^{-3}) - (7.416 \times 10^{-4})$$

$$= 1.9284 \times 10^{-3} \text{ moles} = \text{moles NH}_4\text{Cl}$$

$$\text{4] mass NH}_4\text{Cl} = \# \text{ moles} \times \text{M.M}$$

$$= 1.9284 \times 10^{-3} \times 53.5$$

$$= 0.1032 \text{ g}$$

$$\text{5] NH}_4\text{Cl \%} = \frac{\text{mass NH}_4\text{Cl}}{\text{mass of sample}} \times 100$$

$$= \frac{0.1032}{0.2354} \times 100$$

$$= 43.84 \%$$

$$\begin{aligned}
 \boxed{1} \text{ total moles HCl} &= M_{\text{HCl}} \times V_{\text{HCl}} \\
 &= 0.103 \times 50 \times 10^{-3} \\
 &= 5.15 \times 10^{-3} \text{ moles}
 \end{aligned}$$

$$\begin{aligned}
 \boxed{2} \text{ Moles NaOH (Back titrated)} &= \text{Moles (HCl) un Reacted} \\
 &= M_{\text{NaOH}} \times V_{\text{NaOH}} \xrightarrow{\text{burette}} \\
 &= 0.089 \times 26 \times 10^{-3} \\
 &= 2.314 \times 10^{-3} \text{ moles}
 \end{aligned}$$

$$\begin{aligned}
 \boxed{3} \text{ mole HCl reacted} &= (5.15 \times 10^{-3}) - (2.314 \times 10^{-3}) \\
 &= 2.836 \times 10^{-3}
 \end{aligned}$$

$$\begin{aligned}
 \boxed{4} \text{ moles CaCO}_3 &= \frac{1}{2} \text{ mole HCl reacted} \\
 &= 1.418 \times 10^{-3} \text{ moles}
 \end{aligned}$$

$$\begin{aligned}
 \boxed{5} \text{ mass CaCO}_3 &= \# \text{ mole} \times M.M \\
 &= 1.418 \times 10^{-3} \times 100 \\
 &= 0.1418 \text{ g}
 \end{aligned}$$

$$\begin{aligned}
 \boxed{6} \text{ CaCO}_3 \% &= \frac{\text{mass CaCO}_3}{\text{mass sample}} \times 100 \\
 &= \frac{0.1418}{0.2438} \times 100 = 58.16 \%
 \end{aligned}$$

Exp. 7

Report sheet

Name: Shyma Ibrahim Mohammad (132293)

Section: 4

Unknown no.: 72

1) The concentration of standardized HCl: 0.1 M

2) The composition of Unknown: $\text{Na}_2\text{CO}_3 + \text{NaHCO}_3$

$2V_1 < V_2$

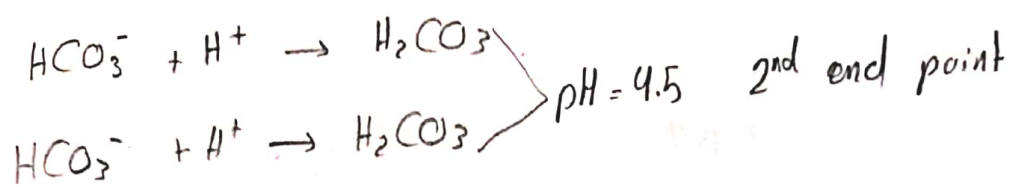
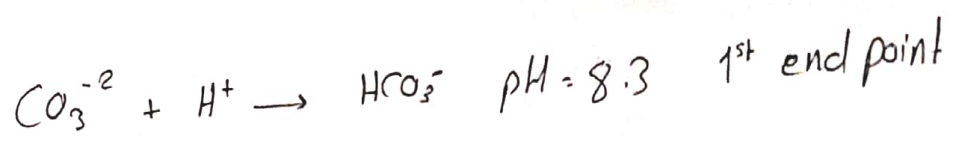
Component 1 is... Na_2CO_3

Component 2 is... NaHCO_3

3) Solved equations:

4) The concentration of Component 1 is... 0.012
~ 0.01

The concentration of Component 2 is... 0.018 40.02



at $V_1 = 3$

$$n \# \text{H}^+ = n \# \text{CO}_3^{2-}$$

$$MV_1 = \quad$$

$$0.1 \times 3 \times 10^{-3} = \quad$$

$$\Rightarrow \boxed{\# \text{CO}_3^{2-} = 3 \times 10^{-4}}$$

at $V_2 = 10.5$

$$\# \text{H}^+ = 2 \# \text{CO}_3^{2-} + \# \text{HCO}_3^-$$

$$MV_2 = 2X + Y$$

$$0.1 \times 10.5 \times 10^{-3} = 2 \times 3 \times 10^{-4} + Y$$

$$\boxed{\# \text{HCO}_3^- = 4.5 \times 10^{-4}}$$

$$M(\text{Na}_2\text{CO}_3) = \frac{\# \text{CO}_3^{2-}}{V} = \frac{3 \times 10^{-4}}{25 \times 10^{-3}} = 0.012 \approx 0.01$$

$$M(\text{NaHCO}_3) = \frac{\# \text{HCO}_3^-}{V} = \frac{4.5 \times 10^{-4}}{25 \times 10^{-3}} = 0.018 \approx 0.02$$

